

ASSIGNMENT No: 6

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TITLE: Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement: Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives: To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory: An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling.

Exercise

1. Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

```
J Vehicle.java > Vehicle > VehicleDetails > display()
1  class Vehicle {
2
3      String name = "Car";
4      String model = "Toyota";
5
6      // Inner Class
7      class VehicleDetails {
8          void display() {
9              System.out.println("Vehicle Name: " + name);
10             System.out.println("Vehicle Model: " + model);
11         }
12     }
13
14     Run | Debug
15     public static void main(String[] args) {
16
17         Vehicle v = new Vehicle();
18
19         // Inner Class Object
20         VehicleDetails vd = v.new VehicleDetails();
21         vd.display();
22
23         // Anonymous Class
24         Runnable action = new Runnable() {
25             public void run() {
26                 System.out.println(x: "Vehicle is Running...");
27             }
28         };
29
30         action.run();
31     }
32 }
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\jdk
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redhat.
Vehicle Name: Car
Vehicle Model: Toyota
Vehicle is Running...
PS C:\Users\arhaa\OneDrive\Desktop\java>
```

2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

```
J fooddelivery.java > FoodDelivery > main(String[])
1  class FoodDelivery {
2
3      String orderId = "FD101";
4      String foodItem = "Pizza";
5
6      // Inner Class
7      class OrderDetails {
8          void display() {
9              System.out.println("Order ID: " + orderId);
10             System.out.println("Food Item: " + foodItem);
11         }
12     }
13
14     Run | Debug
15     public static void main(String[] args) {
16
17         FoodDelivery fd = new FoodDelivery();
18
19         // Inner Class Object
20         OrderDetails od = fd.new OrderDetails();
21         od.display();
22
23         // Anonymous Class for Delivery Status
24         Runnable deliveryStatus = new Runnable() {
25             public void run() {
26                 System.out.println(x: "Order Delivered Successfully!");
27             }
28         };
29
30         deliveryStatus.run();
31     }
32 }
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' %*
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redhat.java\jdt_u
Order ID: FD101
Food Item: Pizza
Order Delivered Successfully!
PS C:\Users\arhaa\OneDrive\Desktop\java>
```